Grade 8 Name_Answer Key________
Unit 12 Date____________________
Lesson 3 Practice Period___________________

For questions 1-3, consider the monomial $(-5dg^4)$ and the algebraic expression $\left(-2d + \frac{12}{5}g^3\right)$.

1. Determine the product of $(-5dg^4)$ and $\left(-2d + \frac{12}{5}g^3\right)$ using an area model.

	$-2d$	$\frac{12}{5}d^3$
$-5dg^4$	$10d^2$	$-12dg^7$

$$= 10d^2 - 12dg^7$$

2. Determine the product of $(-5dg^4)$ and $\left(-2d + \frac{12}{5}g^3\right)$ by applying the distributive property.

$$-5dg^4\left(-2d + \frac{12}{5}g^5\right) = 10d^2g^4 - 12dg^7$$

3. Write another expression equivalent to the product.

$$-5dg^4(-2d) + (-5dg^4)\left(\frac{12}{5}g^3\right)$$

For questions 4-6, consider the monomial $(-1.5x^3y^5)$ and the algebraic expression $(-4x^2 - 3.2y)$.

4. Determine the product of $(-1.5x^3y^5)$ and $(-4x^2 - 3.2y)$ using an area model.

	$-4x^2$	$-3.2y$
$-1.5x^3y^5$	$6x^5y^5$	$+4.8x^3y^6$

$$= 6x^5y^5 + 4.8x^3y^6$$

5. Determine the product of $(-1.5x^3y^5)$ and $(-4x^2 - 3.2y)$ by applying the distributive property.

$$-1.5x^3y^5(-4x^2 - 3.2y)$$

$$= 6x^5y^5 + 4.8x^3y^6$$

6. Write another expression equivalent to the product.

$$(-1.5x^3y^5)(-4x^2) - (-1.5x^3y^5)(3.2y)$$

For questions 7-10, find each product using your preferred method.

7. $-\frac{4}{5}a^3b^2(10b^4 + a^3)$

$-8a^3b^6 - \frac{4}{5}a^6b^2$

8. $20r^5s(0.2s^6 - 4r)$

$4r^5s^7 - 80r^6s$

9. $\left(-\frac{1}{3}h^3 + 7n^4\right)\left(-\frac{7}{8}h^2n^2\right)$

$\frac{7}{24}h^5n^2 - \frac{49}{8}h^2n^6$

10. $-3zb^2\left(8z^5 - \frac{7}{3}b^2\right)$

$-24b^2z^6 + 7b^4z$